

Figure 5: Scree plot

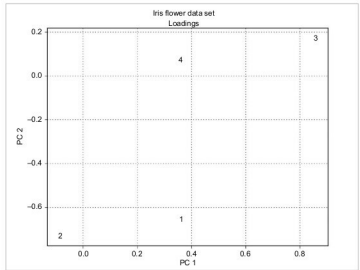


Figure 7: Loadings - PC 1 vs PC 2

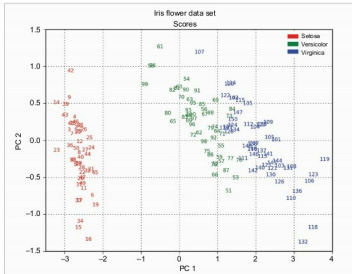


Figure 6: Scores - PC 1 vs PC 2

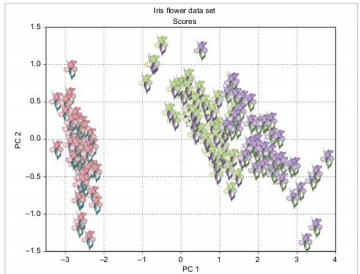


Figure 8: Scores with clipart

50, iris versicolor from row 51 to 100, iris virginica from row 101 to 150). The code is short, as shown below:

```
import scipy.cluster.hierarchy as hca
import xlrd
from pylab import *
rcParams["font.family"]=["DejaVu Sans"]
rcParams["font.size"]=10
w=xlrd.open_workbook("iris.xls").sheet_by_name("Sheet1")
data=array([[w.cell_value(r,c) for c in range(w.ncols)] for r
in range(w.nrows)])
dataS=data-mean(data,axis=0)
o=range(1,w.nrows+1)
y=hca.linkage(dataS,metric="euclidean",method="ward")
hca.dendrogram(y,labels=o,color_threshold=10,truncate_
mode="lastp",p=25)
xticks(rotation=90)
tick_params(top=False,right=False,direction="out")
ylabel("Distance")
```

```
figtext(0.5,0.95,"Iris flower data
set",ha="center",fontsize=12)
figtext(0.5,0.91,"Dendrogram (center, euclidean,
ward)",ha="center",fontsize=10)
savefig("figure.png",format="png",dpi=300)
```

First, the table is read as an array with a nested loop; then, column centring is performed. There are various other scaling techniques, column centring is an example of one of them. The 'S' in *dataS* is for 'scaled'. In this example, the *euclidean* metric and the *ward* linkage method are chosen. Many other metrics are available—for example, *canberra*, *cityblock*, *mahalanobis*, etc. There are also many other linkage methods—for example, *average*, *complete*, *single*, etc. Finally, the dendrogram is plotted as shown in Figure 4. In this example, the clusters are coloured by cutting the dendrogram at a distance equal to 10, using the option *color_threshold*. To enhance its readability, the dendrogram has been condensed a bit using the option *truncate_mode*.